Helm Platform — Phase 4 Code Review

Date: 2026-02-22 **Reviewed by:** Manus AI

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1. Executive Summary

The Phase 4 commit is another **outstanding** delivery. The developer has successfully implemented all three parts of the prompt, integrating live data from Strapi and Mapbox, and delivering the requested UI/UX enhancements. The quality of the code remains exceptionally high, and the project continues to be in a very strong position.

Overall Assessment: Approved

There are no blocking issues. The work is approved to move to Phase 5.

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2. Detailed Findings

2.1. Part 1: Strapi Integration for Signature Experiences

Status: ✅ Complete and Correct

- **Backend:** The new `GET /api/v1/loyalty/experiences` endpoint is implemented perfectly. The fallback to a default list when Strapi is unavailable is a great piece of defensive programming. The redemption validation logic in `redeem_points` is now robust and correctly checks against the live Strapi catalogue.
- **Frontend:** The Crew Loyalty dashboard now correctly fetches and displays the dynamic list of experiences from the new endpoint. The loading and error states are handled gracefully.

2.2. Part 2: Live Mapbox Integration for Helm Dash

Status: ✅ Complete and Correct

- The placeholder map has been successfully replaced with a live `MapboxMap` widget. The use of a `_MapboxView` StatefulWidget to manage the controller and markers is a clean and effective approach.
- The fallback to a placeholder view when no `MAPBOX_ACCESS_TOKEN` is provided is excellent, ensuring the app remains functional for developers without a token.
- The live updating of the delivery vessel's position is working as specified.

2.3. Part 3: UI/UX Enhancements

Status: ✅ Complete and Exceeds Expectations

- **Multi-Image Product Gallery:**
  - The backend schema change from a single `imageUrl` to a `list[str]` of `images` is correct.
  - The frontend implementation is excellent. The `PageView.builder` with a `PageIndicator` is exactly what was requested. The tap-to-expand full-screen gallery using `photo_view` is a great addition that significantly improves the user experience.
- **Markdown Rendering for AI Chat:**
  - The `Text` widget has been successfully replaced with a `MarkdownBody` widget for assistant messages.
  - The styling is clean and well-integrated with the app's theme.
  - The developer correctly ensured that user messages are still rendered as plain text, preventing any potential injection issues.

2.4. Test Coverage

Status: ✅ Excellent

Test coverage remains a key strength of this project. The new backend tests for the Strapi integration (`test_experiences.py`) are comprehensive. The new frontend

widget tests for the product gallery and Markdown rendering are also excellent and cover all key states and interactions.

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3. Recommendations for Phase 5

There are no high-priority issues. The project is now feature-complete with respect to the initial specification. The next phase should focus on finalising the user journey, adding the remaining screens, and preparing for a production release.

- **High Priority:**
    1. **Implement Checkout Flow:** Create the full multi-step checkout flow, including shipping address, payment method selection (Stripe/Afterpay/Laybuy), and order confirmation.
    2. **Implement Voyage Checklists:** Create the UI for the tiered voyage checklists (Day, Coastal, Offshore), allowing users to view, manage, and add missing items to their cart.
- **Medium Priority:**
    1. **User Profile Screen:** Add a user profile screen where users can manage their details, view their order history, and manage their notification settings.
    2. **Onboarding Flow:** Create a simple onboarding flow for new users that guides them through creating their first “My Vessel” profile.